

What is a Passive Attack?

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Monitoring network data without altering system resources.

- Silent data extraction protocols.
- Zero system state alteration.
- Invisible to standard firewalls.

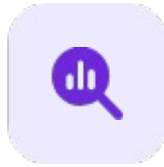
Characteristics of Passive Attacks

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Undetectable

- Leaves zero digital footprint.
- Avoids standard system logs.
- Bypasses intrusion detection.



Information Goal

- Extracts plaintext network secrets.
- Maps internal infrastructure topologies.
- Identifies high-value targets.



Prevention Focus

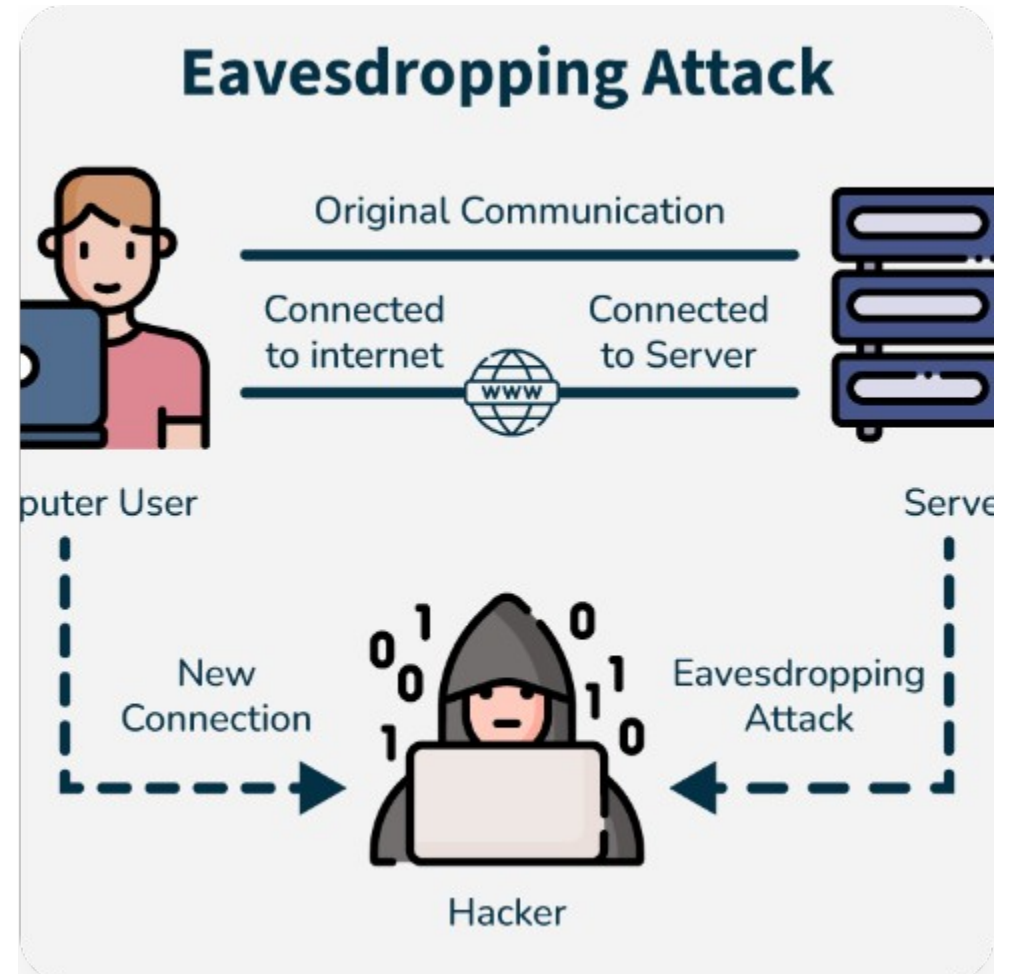
- Requires strong encryption algorithms.
- Demands secure key exchanges.
- Traffic masking is essential.

Passive: Eavesdropping

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Intercepting unencrypted network traffic in real-time.

- Sniffing open Wi-Fi networks.
- Reading plaintext email communications.
- Capturing exposed API keys.
- Monitoring unencrypted messaging apps.



Passive: Traffic Analysis

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Extracting actionable metadata from encrypted payloads.



Identity Mapping

- Maps sender IP addresses.
- Tracks receiver nodes.



Frequency Tracking

- Logs active operational hours.
- Quantifies exact packet bursts.



Length Measurement

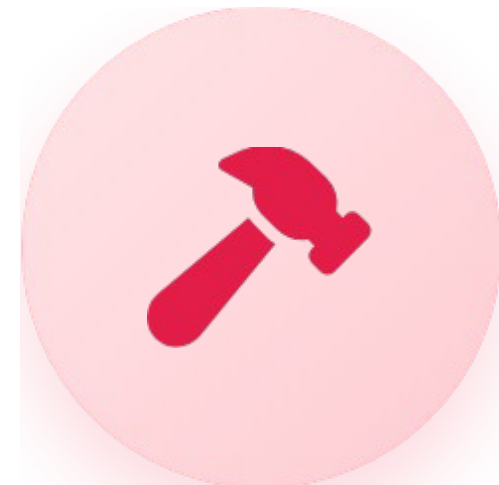
- Guesses encrypted payload types.
- Measures exact file sizes.

What is an Active Attack?

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Modifying network data streams to compromise system integrity.

- Injecting malicious code payloads.
- Deleting critical system logs.
- Triggering immediate security alerts.



Characteristics of Active Attacks

Characteristics of Active Attacks



Destructive

- Destroys core system resources.
- Corrupts critical corporate databases.
- Halts primary business operations.



Detectable

- Generates massive system anomalies.
- Triggers standard firewall alarms.
- Leaves extensive digital forensics.



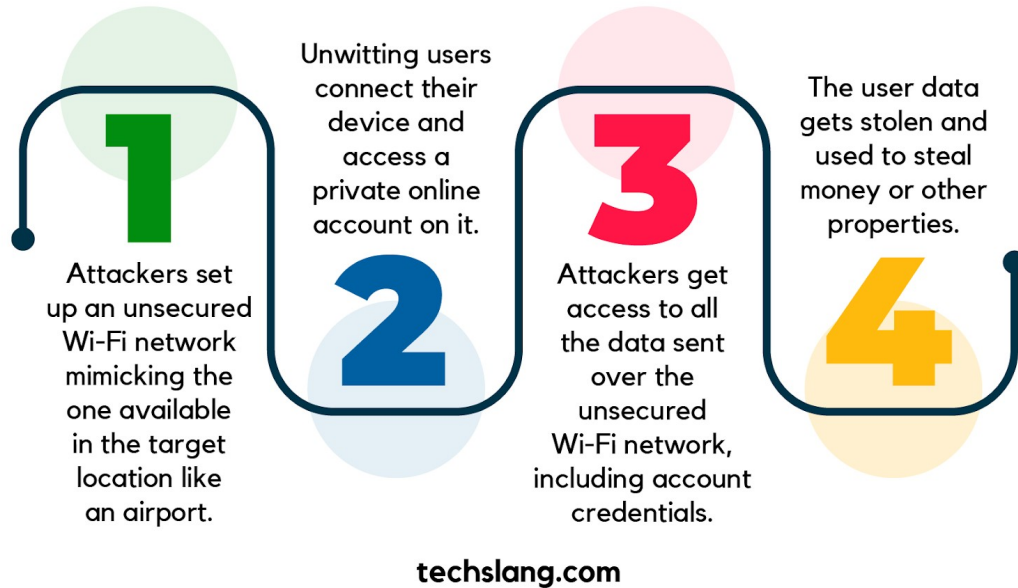
Mitigation Focus

- Requires rapid incident response.
- Demands automated threat isolation.
- Recovery speed dictates survival.

Active: Interception & Modification

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HOW DATA INTERCEPTION ATTACKS WORK



Capturing data, altering contents, and forwarding corrupted payloads.

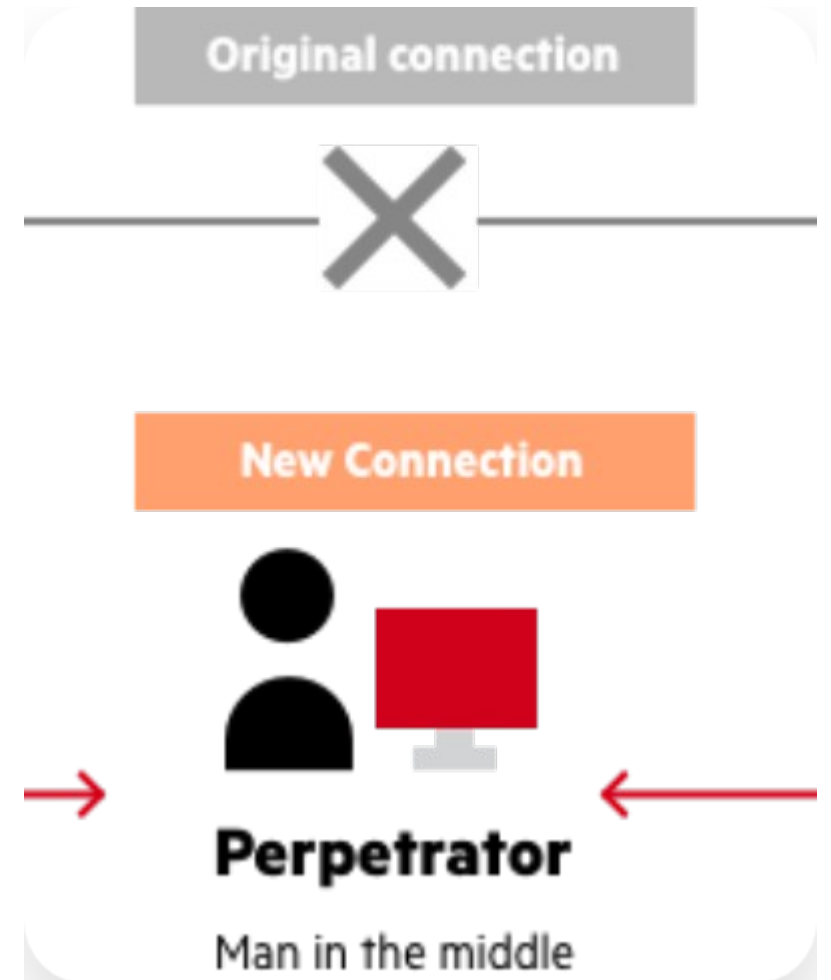
- Breaking original data integrity completely.
- Injecting false system execution commands.
- Rerouting active financial wire transactions.
- Spoofing authorized user identities.

Man-in-the-Middle (MITM)

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The Invisible Intermediary

- Attacker secretly relays all messages.
- Alters live communication in transit.
- Both victims assume absolute privacy.
- Controls the entire conversation flow.



Consequences of MITM

Consequences of MITM



Credential Theft

- Capturing passwords in transit.
- Harvesting multi-factor auth codes.



Session Hijacking

- Stealing active authentication tokens.
- Bypassing secure login portals.



Data Forgery

- Injecting malicious system commands.
- Altering financial routing numbers.

Replay Attacks

Replay Attacks

Malicious repetition of valid network transmissions.

- Capturing valid user authentication tokens.
- Delaying payload transmission indefinitely.
- Resending packets to spoof identity.
- Gaining unauthorized administrative access.



The Replay Attack Sequence

The Replay Attack Sequence

1

Intercept

Capture valid authentication token from network.

2

Store

Hold token securely in malicious proxy server.

3

Resend

Bypass authentication systems entirely via replay.

Defensive Countermeasures

Defensive Countermeasures



Defeating MITM

- Mandatory TLS 1.3 protocol encryption.
- Strict public key certificate pinning.
- Mutual TLS client-server authentication.
- Continuous network anomaly monitoring.



Defeating Replay

- Single-use session authorization tokens.
- Cryptographic random nonces injected.
- Strict packet time-stamping rules.
- Sequence numbering for all messages.

Summary: Passive vs. Active Attacks

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Feature	Passive Attacks	Active Attacks
Primary Goal	Silent information gathering & extraction	System disruption & data corruption
Data Modification	Strictly none; read-only operations	Alters, deletes, or injects data streams
Detectability	Nearly impossible; leaves zero logs	Highly detectable via system anomalies
Core Defense	Prevention (Strong encryption suites)	Mitigation (Intrusion Detection Systems)